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Estimation of frequency components in stator current for the detection of broken rotor bars in induction machines

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ABSTRACT

The limitation of data window length in induction machine broken rotor bar diagnostics is a real challenge in practice. Sideband frequencies which are used as broken rotor bar indicators are very close to the fundamental frequency and have low magnitude. Traditional spectral analysis approach such as Discrete Fourier Transform (DFT) can be inaccurate in these conditions due to its inherent drawbacks such as the requirement of long data window for high resolution and the side lobe leakage in frequency domain. In this paper, a high-resolution spectral analysis technique, Prony Analysis (PA), is proposed for broken rotor bar detection in induction machines. The method is described and demonstrated in detail, validated by experimental data, and compared with DFT. Results clearly indicate the advantages of PA over DFT in terms of maintaining a high resolution with a much shorter window and a better frequency estimate accuracy with the same window length.
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1. Introduction

Induction machines (IM) play a vital role in industry. Internal machine faults such as broken rotor bars are potential hazards to the reliability and safety of industrial operations, and also increase the operational costs. Therefore, induction machine condition monitoring and fault diagnostics has been an important research topic in the last decades, and many of them are with great respect to broken rotor bar detection [1–4].

Motor Current Signature Analysis (MCSA) is a widely adopted technique for on-line diagnostics of induction machine broken rotor bars. Research has found that when broken bars occur in the machine rotor, the induced anomaly of the electromagnetic field will cause two sideband frequency components, $(1 \pm 2s)f$, to appear in the stator current spectrum [5], where f is the supply frequency and s is the machine slip. Identification of these sideband frequencies can be used as a convenient and reliable approach to broken rotor bar fault diagnostics [3].

In an induction machine, the slip s is usually below 10%. If the machine load gets lighter, the slip gets smaller. As a consequence, the two broken rotor bar sideband frequencies become quite close to the fundamental frequency. This appears in the stator current spectrum as two small peaks in the vicinity of a high peak. Combined with low signal to noise ratio, the task of selecting a spectral analysis method is difficult.

Discrete Fourier Transform (DFT) is a classical technique for spectral analysis. Fast Fourier Transform (FFT), which is a fast computation algorithm of DFT, has been previously adopted in the implementation of MCSA [5,3]. However, inherent disadvantages of using DFT limit its applicability in practice. The two broken rotor bar sideband frequency components are very close to the fundamental frequency when the machine load is light. This requires the data window used for DFT to be long for sufficient frequency resolution [6]. Another major disadvantage of DFT is that the impact of side wiggles (Gibbs oscillations). The implicit windowing process when using DFT causes side lobe leakage in spectral domain, obscuring and distorting other spectral response in its vicinity [7]. There are also other limitations such as that the time domain noise in the signal is distributed uniformly by DFT in the frequency domain,

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which limits the certainty of computing frequency widths, magnitudes, and phases [8] and, it is also known that DFT may cause spurious spectral components in the spectrum, which will confuse the detection on desired frequency components. Additionally, DFT requires the values of the frequencies in the target signal to be constant.

These limitations of DFT can be particularly troublesome in real induction machine condition monitoring situations. Short data records are usually required because the instability of machine load condition may cause time-varying broken rotor bar sideband frequencies. However, on the other hand, a high resolution is required to observe the two broken rotor bar sideband frequencies when the machine is operating with light load. This means longer data acquisition time. Moreover, in practice, sometimes only restricted data records are available. This also makes enlarging the data window to obtain a high resolution impossible. Thus, trade-offs among leakage suppression, resolution and tracking changes are difficult to fulfill. The detection of broken rotor bars using DFT can be difficult and unauthentic. Failures have been reported in [5,9]. In most applications reported in the literature, where DFT is used for the signal processing stage of MCSA, the machine load is usually fixed at near full load.

Due to those limitations of DFT, other methods for spectral analysis become potential options to the detection of broken rotor bars. Prony Analysis (PA) is a high-resolution spectral analysis method developed on the basis of the original work of the French mathematician, Gaspard de Prony [10]. It is able to achieve high frequency resolution and estimation accuracy using very short data acquisition windows. Such technique overcomes the drawbacks of DFT and has high value in practical implementation in light and variable machine load conditions.

In this paper, firstly, the original Prony's method and the Least Squares (LS) Prony, which is used for dealing with real signals, are reviewed. It is known that the original Prony's method is sensitive to noise [8]. The LS Prony method deals with practical situations but cannot provide satisfied outcomes since it fits the model to any noise present in the data [10]. Therefore, the Iteratively Reweighted Least Squares (IRLS) Prony [11] is then proposed to obtain improved results in the broken rotor diagnosis application. The term PA in this paper refers to the IRLS Prony method, except otherwise indicated. In the following sections, results obtained by PA using both simulated and experimental data are presented and compared with that obtained by DFT, clearly demonstrating the superiority of PA over DFT in terms of using much shorter data windows to achieve high resolutions and obtaining higher precision in frequency estimation with the same length of window. In the end, influential factors to the PA are analyzed.

2. Prony Analysis

2.1. Theory of the Prony's method

The original Prony method seeks to fit a deterministic exponential model to equally spaced data points. It was discussed in detail by Marple [10] and Therrien [12]. Here we will give a brief review of this technique. Assuming signal data $x[n]$ has N complex samples $x[1],\ldots,x[N]$, the Prony method will fit the data with a sum of q complex exponential functions

$$\hat{x}[n] = \sum_{k=1}^{q} A_k \exp[(\alpha_k + j2\pi f_k)(n-1)T_s + j\varphi_k] \tag{1}$$

for $n = 1,2,\ldots,N$ and $k = 1,2,\ldots,q$, where A_k is the amplitude of the complex exponential, α_k is the damping coefficient in s^{-1}, f_k is the sinusoidal frequency in Hz, φ_k is the initial phase in radians, and T_s is the sampling interval in seconds.

The objective is to estimate the frequencies f_k, damping factors α_k, amplitudes A_k and phases φ_k. If these function coefficients are determined correctly, then the plot of the estimation of the signal within the data window should fit the original signal with a high degree of accuracy.

Since only real signals are considered, the signal poles $\exp(\alpha_k + j2\pi f_k)$ must appear in complex conjugate pairs. Thus the q is always assumed to be even for convenience. Then, Eq. (1) can be expressed in form of

$$\hat{x}[n] = \sum_{k=1}^{q} h_k z_k^{n-1} \tag{2}$$

where h_k and z_k are complex parameters defined as

$$h_k = A_k \exp(j\varphi_k)$$

and

$$z_k = \exp[(\alpha_k + j2\pi f_k)T_s]$$

The fitting of a designated signal using (2) is a difficult nonlinear problem as sums involving exponentials must be solved for z_k [10]. No analytic solution is available.

Prony's method addresses this problem by determining the z_k elements separately and then considering Eq. (2) as a set of linear simultaneous equations to solve for h_k. The key of the Prony method is in the fact that to see Eq. (2) as the solution to a homogeneous linear difference equation with constant coefficients. Then the coefficients can be identified by solving the linear prediction equation, as demonstrated below. A polynomial with roots z_k can be formed accordingly

$$\phi(z) = \prod_{k=1}^{q}(z - z_k) = \sum_{m=0}^{q} a_m z^{q-m} \tag{3}$$

The linear difference equation whose homogeneous solution is given by Eq. (3) is

$$\sum_{m=0}^{q} a_m x[n-m] = 0 \tag{4}$$

with complex coefficients a_m such that $a_0 = 1$.

The original Prony method assumes that the number of available data samples is equal to the unknown parameters, so the difference equation is valid for $n = q+1,\ldots,2q$. The coefficients a_m form a linear predictive relationship among the available samples and the relationship can be then expressed as the $q \times q$ Toeplitz structure matrix equation [10]

$$\begin{bmatrix} x[q] & x[q-1] & \cdots & x[1] \\ x[q+1] & x[q] & \cdots & x[2] \\ \vdots & \vdots & \ddots & \vdots \\ x[2q-1] & x[2q-2] & \cdots & x[q] \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_q \end{bmatrix} = -\begin{bmatrix} x[q+1] \\ x[q+2] \\ \vdots \\ x[2q] \end{bmatrix} \tag{5}$$

By solving the matrix equation, the a_m coefficients, which are the function of z_k, can be determined.

Next, the roots z_k of Eq. (3) can be determined by polynomial factoring. The damping factor α_k and the sinusoidal frequency f_k can be obtained from z_k by using the relationships [10]

$$\alpha_k = \ln |z_k|/T_s \tag{6}$$

and

$$f_k = \tan^{-1}[Im\{z_k\}/Re\{z_k\}]/2\pi T_s \tag{7}$$

Finally, these roots are used to obtain the complex parameter h_k in Eq. (2). The amplitude A_k and initial phase φ_k are determined from h_k by using the relationships

$$A_k = |h_k| \tag{8}$$

and

$$\varphi_k = \tan^{-1}[Im\{h_k\}/Re\{h_k\}] \tag{9}$$

To sum up, the Prony method consists of three steps [10]:

Step 1 Determine the linear prediction parameters that fit the observed data. This step is undertaken by solving Eq. (5) for the coefficients a_m.

Step 2 Find roots of the characteristic polynomial formed from the linear prediction coefficients and determine the estimates of the damping factor and frequency of each of the exponential terms. This step consists of polynomial factoring Eq. (3) and solving Eqs. (6) and (7).

Step 3 Solve the original set of linear equations to yield the estimates of the exponential amplitudes and sinusoidal initial phases. This step is to solve the original matrix equation (2), where the matrix of the time-indexed z elements has a Vandermonde structure.

2.2. Prony analysis for practical problems with noisy signals

In this subsection the extended LS Prony method and the IRLS Prony method are introduced.

It should be noticed that in the original Prony's method there is no noise model. This means that the actual noise present in the data will be approximated entirely by complex exponentials, leaving an un-modeled residual energy which manifests itself as parameter estimation errors [13]. It is because of this, the performance of the original Prony method is unstable if there is noise in the signal data.

However, in practice, acquired signal data is always embedded in noise. The Eq. (2) should be modified as the following form for noise corrupted signals [4]:

$$x[n] = \sum_{k=1}^{q} h_k z_k^{n-1} + \epsilon[n] \tag{10}$$

where $\epsilon[n]$ is known as the exponential approximation error and noise which is assumed to be Gaussian distributed and white. If the noise present in the signal is not white then standard filtering methods can be used to whiten the signal so that this model applies, too.

The classical Prony method models a sequence of $2q$ observations sampled at even time intervals by q exponential functions at the most. In practice, there are usually more data points than the minimum number of samples needed to fit a model of order q. To deal with practical situations, appropriate least squares procedures and Singular Value Decomposition (SVD) are employed in the first and third steps of the original Prony method [14], and this is called the extended Least Squares Prony method. The goal of the algorithm is to minimize the total squared error over all sampled data with respect to the complex parameters and the number of exponents.

As mentioned in Section 2.1 that the minimization of the norm of the exponential error $\epsilon[n]$ is a difficult nonlinear problem. The extended LS Prony method employs a suboptimum solution that predicts the linear prediction approximation error $e[n]$ instead of the exponential error $\epsilon[n]$ over data $1 \leqslant n \leqslant N$. In an over-determined approach, N data points, where $N > 2q$, are utilized to compose the linear prediction equations. Thus, the linear difference equation Eq. (4) should be modified to [10]:

$$\sum_{m=0}^{q} a_m (x[n-m] + \epsilon[n-m]) = 0 \tag{11}$$

LS Prony method ignores the past noise values, then the Eq. (11) becomes

$$\sum_{m=0}^{q} a_m x[n-m] = e[n] \tag{12}$$

which actually defines a forward linear prediction error equation. Thus each a_m terms are linear prediction parameters and are selected to minimize the linear prediction total squared error $\sum_{n=q+1}^{N} |e[n]|^2$. The minimization can be solved by using the covariance method, or alternatively, by using the SVD for efficient computation of the pseudo-inverse and projection matrices [12,14].

The third step of the original Prony method also switches to a linear least square procedure. The complex-valued $q \times q$ matrix normal equation

$$[\mathbf{Z}^H \mathbf{Z}]\mathbf{h} = [\mathbf{Z}^H \mathbf{x}] \tag{13}$$

can be yielded from Eq. (10). The equation components, which are the $N \times q$ matrix $\mathbf{Z}$, the $q \times 1$ column vectors of $\mathbf{h}$, and the $N \times 1$ column vectors of data $\mathbf{x}$ are defined as

$$\mathbf{Z} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ z_1 & z_2 & \cdots & z_q \\ \vdots & \vdots & \ddots & \vdots \\ z_1^{N-1} & z_2^{N-1} & \cdots & z_q^{N-1} \end{bmatrix}$$

$$\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_q \end{bmatrix}, \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} x[1] \\ x[2] \\ \vdots \\ x[N] \end{bmatrix}$$

The superscript H means the matrix complex conjugate transposition, so that $\mathbf{Z}^H\mathbf{Z}$ forms a $q \times q$ Hermitian matrix. The four parameters of Prony model can be determined in the same way by using Eqs. (6)–(9).

The LS Prony can deal with practical situations but it is still inconsistent in providing unbiased parameter estimates as the number of sampled points increases [15,16]. It is considered as a "suboptimum" solution as this approach does not make a separate estimation of the noise process, but fits the exponentials to any noise present in the data [10]. However, this can be improved significantly by the IRLS Prony method.

The IRLS Prony method was developed for the identification of the resonant-grounded system parameters based on fault records of a power system [11]. The major improvement to the LS Prony method is that it solve the weighted least squares problem

$$\min\{\boldsymbol{\epsilon}^T\mathbf{W}(\mathbf{a})\boldsymbol{\epsilon}\} \tag{14}$$

with respect to $\mathbf{a}$, where the superscript "T" indicates matrix transpose. The weight matrix $\mathbf{W}(\mathbf{a})$ is the inverted covariance matrix for error $\boldsymbol{\epsilon}[n]$ at each data point in the first step of Prony method. It iteratively minimizes the total squared error, so that to filter out noise more efficiently. Accurate parameter identification has been achieved as a result.

We start the IRLS Prony method by the model given by Eq. (10), noting that the measurement errors are assumed to be independent and normally distributed. Taking into account that the error-free signal satisfies exactly the difference equation (4), when substitute the error-free data with real signal data, the difference equation (11) is satisfied for each sample in the data window spanning N samples. It may be expressed in the matrix notation [11]

$$\mathbf{Xa} + \mathbf{b} + \mathbf{D}(\mathbf{a})\boldsymbol{\epsilon} = 0 \tag{15}$$

where $\mathbf{a}^T = [a_1, a_2, \ldots, a_q]$ is the $q \times 1$ column vectors of difference equation coefficients,

$\boldsymbol{\epsilon}^T = [\epsilon[1], \epsilon[2], \ldots, \epsilon[N]]$ is the $N \times 1$ column vectors of error component on each sample,

$\mathbf{b}^T = [x[q+1]\ x[q+2]\ \ldots\ x[N]]$ is the $(N-q) \times 1$ column vectors of data,

$$\mathbf{X} = \begin{bmatrix} x[q] & x[q-1] & \cdots & x[1] \\ x[q+1] & x[q] & \cdots & x[2] \\ \vdots & \vdots & \ddots & \vdots \\ x[N-1] & x[N-2] & \cdots & x[N-q] \end{bmatrix} \text{ is the}$$

$(N-q)q$ data matrix, and

$$\mathbf{D} = \begin{bmatrix} a_q & a_{q-1} & \cdots & a_1 & 1 & 0 & \cdots & 0 \\ 0 & a_q & a_{q-1} & \cdots & a_1 & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \cdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_q & a_{q-1} & \cdots & a_1 & 1 \end{bmatrix} \text{ is the}$$

$(N-q)N$ coefficients matrix for errors

In order to minimize the sum of squared error over the available data, the error $\boldsymbol{\epsilon}$ is expressed by rearranging Eq. (15), yielding

$$\boldsymbol{\epsilon} = -\mathbf{D}(\mathbf{a})^+(\mathbf{Xa} + \mathbf{b}) \tag{16}$$

where '+' indicates the matrix pseudoinversion. Thus, the optimal estimates of the difference coefficients a_m correspond to the minimum of the error norm $\boldsymbol{\epsilon}^T\boldsymbol{\epsilon}$. This is the nonlinear problem addressed by Eq. (14) and can be formulated in terms of IRLS, that is, $\mathbf{a}$ minimizes

$$\min_a\{(\mathbf{Xa} + \mathbf{b})^T\mathbf{W}(\mathbf{a})(\mathbf{Xa} + \mathbf{b})\} \tag{17}$$

where $\mathbf{W}(\mathbf{a}) = [\mathbf{D}(\mathbf{a})\mathbf{D}(\mathbf{a})^T]^{-1}$ is a real symmetric positive definite weighting matrix. The Eq. (17) is solved with standard weighted least squares algorithm [17].

One aspect of the IRLS Prony method that needs to be addressed to attention is that because the elements of the weighting matrix $\mathbf{W}(\mathbf{a})$ depend on the unknown parameters, it is essential to apply an iterative scheme using the estimates obtained at the previous iteration. Thus, results of the LS Prony method are used here as the initial values of the iteration process. Then an iteratively reweighting process is the next step based on error residue criteria and the iteration count [11].

3. Data preprocessing

3.1. Prefiltering

Prefiltering is a technique commonly employed prior to Digital Signal Processing (DSP) procedures to gain improved results. For all DFT applications in this chapter, signal data is processed by a Hanning window before applying the FFT algorithm, in order to decrease the spectral leakage effect and to shape the signal spectrum.

The objective of prefiltering the signal prior to applying PA is to attenuate the noise and undesired frequency components and to separate the broken rotor bar sideband frequency components in the stator current spectrum. By doing so, the number of frequency components in the current signal becomes known and the performance of PA can be improved significantly [13,18,19]. Therefore, a finite-duration impulse response (FIR) bandpass filter is designed to have a narrow passband of a few hertz, according to the distance between sidebands and the fundamental frequency, and to center at the fundamental frequency. Normally, the width of the passband is limited within 10 Hz. This prefiltering process should be implemented in digital filtering devices continuously prior to the data being sent for PA in on-line diagnostics applications. When the induction machine is operating with more than one broken rotor bar under a lighter load condition, widths of both the passband and the transition are required to be narrower, since the two sideband frequency components are closer to the fundamental frequency and their harmonics will appear nearby [20]. Additionally, use of a lower sampling frequency can reduce the filter order without compromising much of the accuracy of the PA.

The magnitude response of one possibility of the filter is plotted in Fig. 1. This filter has been used during current acquisition when the machine operates with relatively higher load. The specification of this FIR filter is as following:

Lower stopband edge: 37 Hz, attenuation: 60 dB.
Lower passband edge: 41 Hz, ripple: 1 dB.
Upper passband edge: 59 Hz, ripple: 1 dB.
Upper stopband edge: 63 Hz, attenuation: 60 dB.

3.2. Removing the constant offset

In recorded real data, there is such a concern that a DC component may be introduced to the signal by the electronic devices that are used in the test. The DC component can result in significant errors in the PA result. The bias causes components that will tend toward a zero frequency. Any linear trend and the signal mean existing in the data need to be removed before applying PA [8,21].

3.3. Downsampling

PA involves solving of over-determined linear equations and rooting of high-order polynomials. Both of them are computationally intensive operations. The iterative algorithm is the recurrence of these processes. Thus, the amount of data used in the algorithm can be a great concern in practice as a large number of data points increases the complexity of the equations and demands more computational effort and longer computational time. Downsampling the data signal can effectively reduce the number of data points and the effort of computation. Therefore, a downsampling process is sometimes desired for the benefit of computational efficiency and the performance of PA, when the original sampling frequency is high. To avoid aliasing, anti-aliasing low pass filter should be implemented before downsampling.

4. Simulation tests

Simulated stator current data of induction machines with broken rotor bar faults is used for testing in this section. Details of the induction machine model and the description of the simulation can be found in [20]. A 5.5 kW induction machine with four poles and 28 total rotor bars is simulated. The supply is 50 Hz three-phase voltages with rated amplitude. The noise corrupted stator current data is obtained by adding normally (Gaussian) distributed random signals to the machine input voltages. The noise level in the simulation signals is determined as approximated −80 dB to the supply frequency. This value is chosen since the noise level present in measured signals from laboratory experiments described in the next section for validating PA is observed to be always between −80 dB and −90 dB to the fundamental frequency.

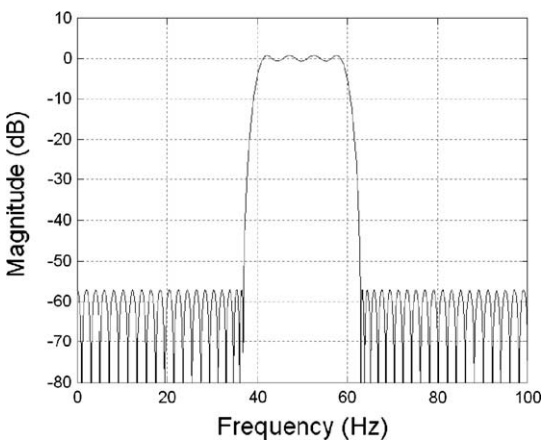

Fig. 1. The magnitude response of the FIR equiripple bandpass filter.

4.1. Prony Prediction

While the number of broken rotor bars increases, the anomaly of the flux linkage within the motor aggravates consequently, which will cause a higher degree of current distortion. Results of the PA prediction for the stator current data inside the window and immediate after the window are plotted in Figs. 2–5a, together with the simulated current signals for comparison. The fitting error at each data point, and the Mean Absolute Error (MAE) of the fitting both inside and after the window are presented. The MAE is calculated as the mean of the absolute error over the whole length of the designated data segment and given as

$$MAE_{fitting} = \frac{\sum_{n=1}^{N} |\epsilon[n]|}{N} \tag{18}$$

where $\epsilon[n]$ is the error calculated on each sampled data point and N is the total number of the data points.

The current signal is sampled at the frequency of 1000 Hz for 0.5 s. Therefore, the window length used for PA is 500 samples (half of the displaying waveform length). The left halves of Figs. 2–5a (up to 0.5 s) reveal the results of PA prediction inside the window for fitting the model, and the right halves of these figures show the PA predictions for data outside the window. The prediction waveforms are obtained by plotting the signal data with parameters gained from the fitting procedure. The induction machine is loaded with full load. The number of broken rotor bars are varied as 0, 3, 5 and 8, where 0 indicates the healthy status of the induction machine and eight is close to one-third of the total number of rotor bars. The induction machine will break down if the number of broken rotor bars reaches, or exceeds one-third of the total number of rotor bars.

The numerical result of PA for Figs. 2–5 is displayed in Table 1. As a linear prediction, Prony method estimates the information of frequency, amplitude, damping factor and phase within a designated signal and tries to fit a model to the signal. Only the value and amplitude of the frequency components are the parameters of interest to induction machine broken rotor bar diagnostics. The damping factor may be used associated with the amplitude to eliminate ambiguous results in the case of using a model order higher than the actual number of signal poles.

It can be observed from the comparison that the more defective rotor bars there are, the more severely the stator

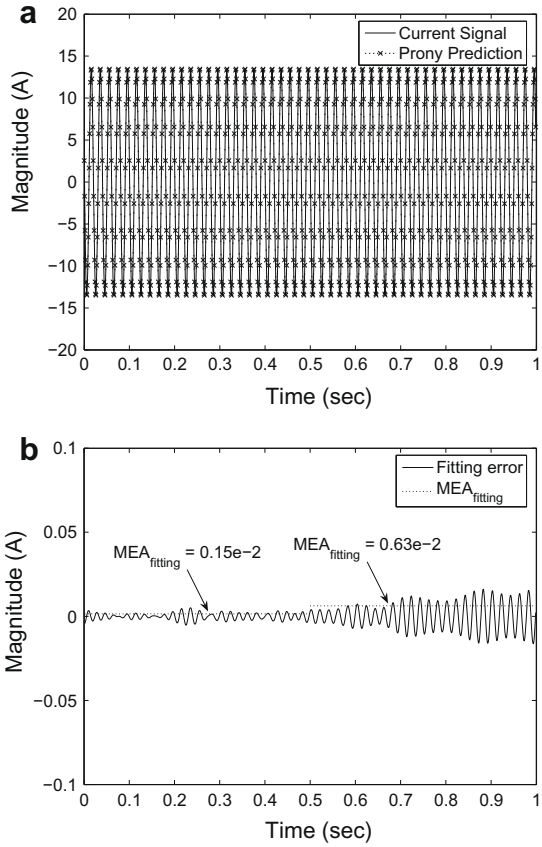

Fig. 2. Comparison of the current waveforms between PA prediction with the simulated current signal inside and outside the window for no broken rotor bar case. (a) PA prediction result inside and immediate after the window. (b) Difference between the current signal and the PA prediction.

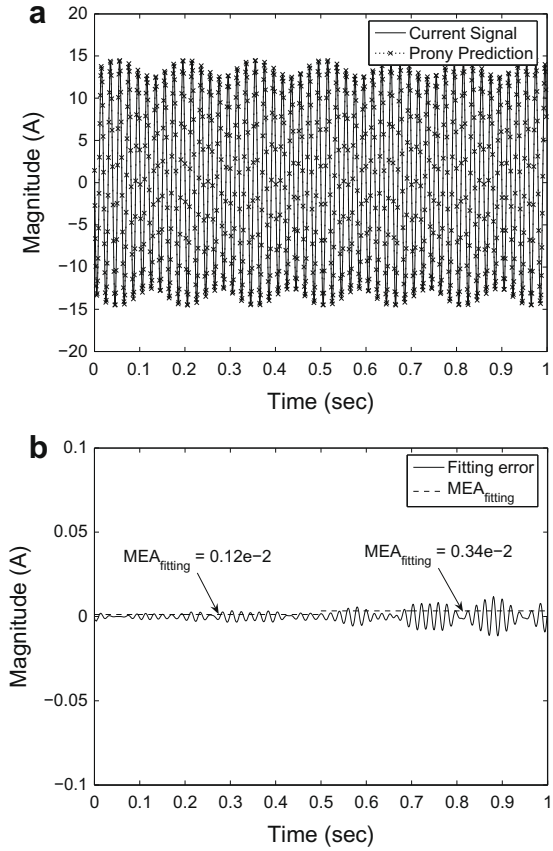

Fig. 3. Comparison of the current waveforms between PA prediction with the simulated current signal inside and outside the window for three broken rotor bar case. (a) PA prediction result inside and immediate after the window. (b) Difference between the current signal and the PA prediction.

current is modulated. It is also shown explicitly that the PA prediction fits both the data within and after the window with acceptable accuracy in all cases. All fundamental and sideband frequency components have been estimated accurately, shown in the table comparing with the true frequency values. The true values of the two sideband frequencies are calculated from the rotor slip using equation $(1 \pm 2s)f$. The result is sufficient to prove the credibility of PA as it does not only exactly model the available data but also well predicts the future data.

4.2. Comparisons between DFT and PA

4.2.1. Window length

It is known that the frequency resolution of DFT, which indicates the capability of distinguishing neighboring frequency components, lies solely on the length of sampling time, or the length of data window with a given sampling frequency. It is because of this, the frequency resolution is a major problem when using DFT, especially in the application of induction machine broken rotor bar diagnosis where due to restrictions that the window length cannot be enlarged as desired.

This disadvantage is demonstrated as follows. Figs. 6–9 present examples of the stator current spectra obtained by DFT using windows of different lengths. The induction machine is simulated under full load. A test case with two broken rotor bars is chosen for demonstration. The sampling frequency is 1000 Hz and the signal data is processed through a Hanning window before applying FFT.

In Fig. 6, a window of 5000 data points is used, which requires a sampling time of 5 s and provides a frequency resolution of 0.2 Hz. The two sideband frequencies are observed distinctly in the spectrum, noticing the true values of the lower and higher sideband frequency components are calculated as 43.72 Hz and 56.28 Hz, respectively, and the frequency values given by DFT are 43.80 Hz and 56.20 Hz. If a shorter data window, for example that of 1000 data points is used, the two frequencies are still visible but with quite a low definition, as shown in Fig. 7. However, Fig. 8 shows when the window size is reduced to 500 samples, DFT fails to distinguish the two broken rotor bar sideband frequencies.

This disadvantage can be worsened if the machine load becomes lighter, as the data window required for DFT will become much longer. Fig. 9 shows the spectral result of using a window of 2000 data points and a same sampling frequency for the same machine as above but operating under 25% of full load. It can be seen the two sideband

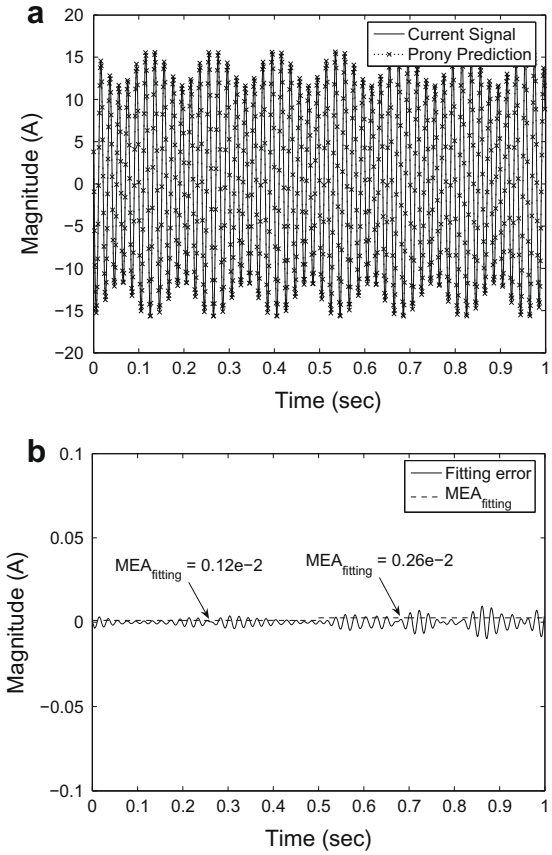

Fig. 4. Comparison of the current waveforms between PA prediction with the simulated current signal inside and outside the window for five broken rotor bar case. (a) PA prediction result inside and immediate after the window. (b) Difference between the current signal and the PA prediction.

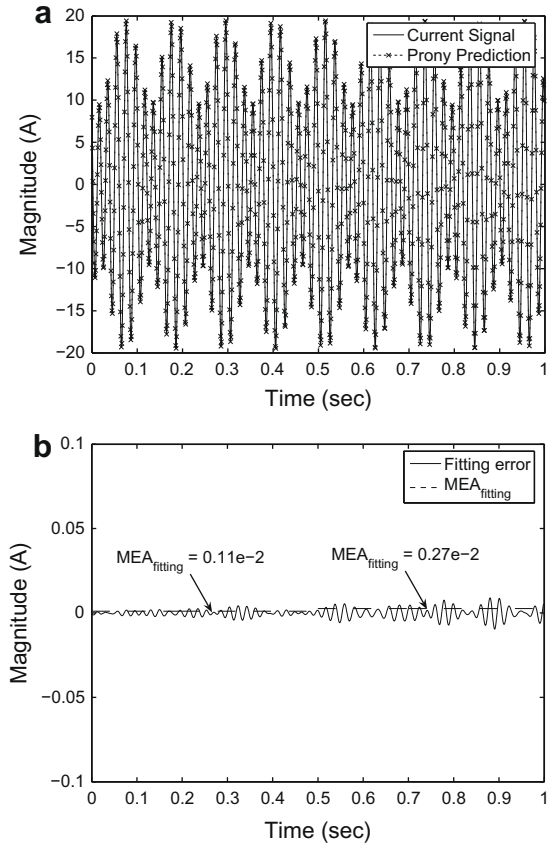

Fig. 5. Comparison of the current waveforms between PA prediction with the simulated current signal inside and outside the window for eight broken rotor bar case. (a) PA prediction result inside and immediate after the window. (b) Difference between the current signal and the PA prediction.

frequency components have merged into the fundamental frequency already and are not able to be observed. If random time-varying load applies, the values of the two sideband frequencies will also be time-varying. The smearing effect will happen typically in the spectrum obtained by the DFT, resulting in failures of detecting broken rotor bars [5].

For the convenience of comparison, minimum window lengths are introduced as the threshold of the window length requirement for both PA and DFT. It is defined as the shortest data window required in order to provide a sufficient degree of accuracy in frequency estimation. Here, this criterion of accuracy is defined as the unitary frequency error, given by

$$UE = \frac{|f_{est} - f_{true}| \times 100}{f_{true}}\% \tag{19}$$

where f_{est} denotes the estimated frequency value whilst f_{true} indicates the true frequency value calculated from motor slip. The minimum or threshold window length is then defined as the shortest window needed for either PA or DFT to maintain the unitary error UE of frequency estimation within 0.3%. It also should be kept in mind that the exact value of each threshold is dependent on its particular condition. It may vary when condition changes.

As an example, a number of simulations have been conducted and the results reveal that for the IM with two broken rotor bars operating under full load, when using a sampling frequency of 1000 Hz, DFT requires a minimum data window with the length of 800 data points (frequency resolution: 1.25 Hz). This requirement is elevated to a 2600 points window (frequency resolution: 0.38 Hz) when there is 25% of full load and up to a 6000 points window (frequency resolution: 0.17 Hz) when the load is only 10% of full load. If a lighter load condition applies, an even longer sampling time is required consequently. These limitations make the accuracy of the broken rotor bar detection in induction machine suffer considerably from machine load variations or lack of data records.

However, there is no such a restriction for PA. To generalize the ability of maintaining a necessary frequency resolution for both PA and DFT with respect to the data window length, Tables 2–5 list the true and estimated values of sideband frequency components obtained by using windows of the minimum lengths for broken rotor bar detection. The number of broken rotor bars and the load condition are varied. The sampling frequency used in these

Table 1
Numerical PA results of the stator current of the IM with various numbers of broken rotor bars operating under full load condition using a data window of 500 samples with the sampling frequency of 1000 Hz.

Number of broken bars	True value		PA						$MEA_{fitting}$ inside the window ($\times 10^{-2}$)	$MEA_{fitting}$ outside the window ($\times 10^{-2}$)
	$(1-2s)f$ (Hz)	$(1+2s)f$ (Hz)	$(1-2s)f$ (Hz)	Amplitude (A)	$(1+2s)f$ (Hz)	Amplitude (A)	Fundamental freqency (Hz)	Amplitude (A)		
0	N/A	N/A	N/A	N/A	N/A	N/A	50.00	6.79	0.12	2.08
1	43.94	56.06	43.94	0.19	56.04	0.05	50.00	6.79	0.52	2.19
2	43.72	56.28	43.72	0.39	56.29	0.09	50.00	6.80	0.58	2.60
3	43.46	56.54	43.46	0.61	56.55	0.13	50.00	6.81	0.45	2.32
4	43.15	56.85	43.15	0.85	56.85	0.17	50.00	6.82	0.58	3.09
5	42.79	57.21	42.79	1.18	57.22	0.21	50.00	6.87	0.29	1.24
6	42.33	57.67	42.33	1.73	57.66	0.26	50.00	6.94	0.27	0.93
7	41.76	58.24	41.76	2.19	58.24	0.28	50.00	7.04	0.39	1.64
8	40.97	59.03	40.96	2.73	59.04	0.29	50.00	7.24	0.24	1.00

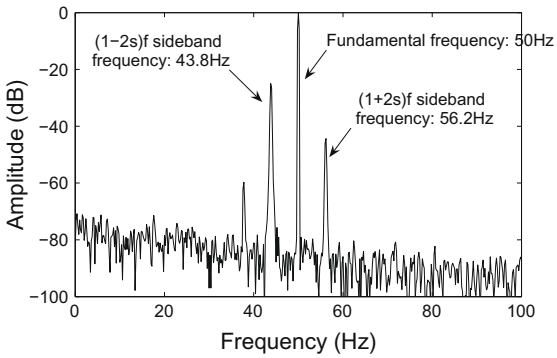

Fig. 6. DFT spectrum of the current signal of the IM with two broken rotor bars operating under full load. The data window length is 5000 samples using a sampling frequency of 1000 Hz.

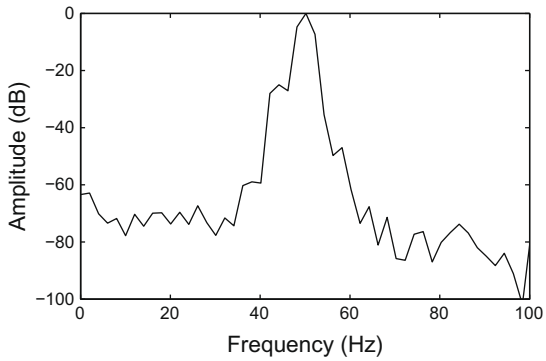

Fig. 8. DFT spectrum of the current signal of the IM with two broken rotor bars operating under full load. The data window length is 500 samples using a sampling frequency of 1000 Hz.

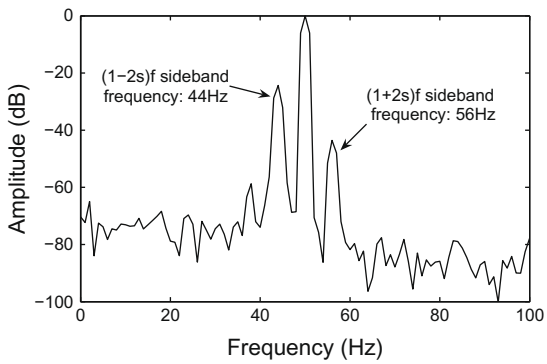

Fig. 7. DFT spectrum of the current signal of the IM with two broken rotor bars operating under full load. The data window length is 1000 samples using a sampling frequency of 1000 Hz.

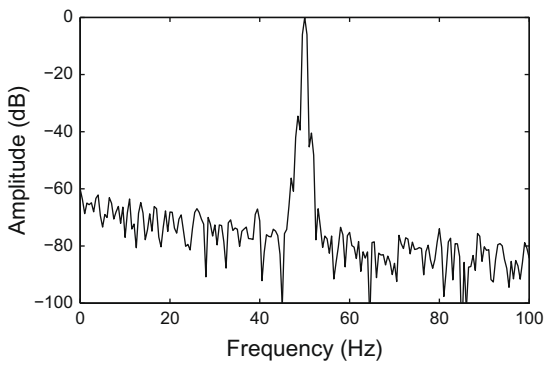

Fig. 9. DFT spectrum of the current signal of the IM with two broken rotor bars operating under 25% of full load. The data window length is 2000 samples using a sampling frequency of 1000 Hz.

simulations is 1000 Hz. The stator currents used for both approaches are the same one in each case.

In Table 2, the result shows that when using even a window size as small as only 40 samples, PA is still able to estimate the values of the sideband frequency components, whereas in the case of using DFT, windows which are more than 20 times longer are required. A similar result is also observed for other load conditions. Besides that the load condition affects considerably the values of sideband frequencies, it is noticed from the table that the number of broken rotor bars also has a small influence on them. This is because the increase of resistance on the rotor due to fractured rotor bars can be treated as a small increase of load. When the number of broken rotor bars rises, the two sideband frequencies are slightly more apart from the fundamental frequency. It also can be observed that the minimum window length requirement for PA decreases as the number of broken rotor bars increases.

Table 2
PA and DFT results of the $(1 \pm 2s)f$ sideband frequencies using the minimum window lengths with 1000 Hz sampling frequency for the IM operating under full load condition and with various numbers of broken rotor bars.

Full load	$(1-2s)f$ (Hz)			$(1+2s)f$ (Hz)			Min. window length (number of samples)	
Number of broken bars	True value	DFT	PA	True value	DFT	PA	DFT	PA
1	43.94	44.00	43.97	56.06	56.00	56.06	1000	40
2	43.72	43.75	43.71	56.28	56.25	56.19	800	40
3	43.46	43.33	43.46	56.54	56.67	56.57	900	40
4	43.16	43.08	43.16	56.84	56.92	56.83	1300	40
5	42.78	42.86	42.78	57.22	57.14	57.25	900	40
6	42.33	42.22	42.33	57.67	57.78	57.67	900	40
7	41.76	41.82	41.76	58.24	58.18	58.25	1100	40
8	40.96	41.00	40.97	59.04	59.00	59.02	1000	40

Table 3
PA and DFT results of the $(1 \pm 2s)f$ sideband frequencies using the minimum window lengths with 1000 Hz sampling frequency for the IM operating under 75% load condition and with various numbers of broken rotor bars.

75% load	$(1-2s)f$ (Hz)			$(1+2s)f$ (Hz)			Min. window length (number of samples)	
Number of broken bars	True value	DFT	PA	True value	DFT	PA	DFT	PA
1	45.62	45.71	45.73	54.38	54.29	54.34	1400	120
2	45.46	45.45	45.43	54.54	54.55	54.42	1100	100
3	45.27	45.38	45.19	54.73	54.62	54.58	1300	100
4	45.06	45.00	45.10	54.94	55.00	55.02	1000	100
5	44.80	44.67	44.80	55.20	55.33	55.20	1500	100
6	44.48	44.55	44.38	55.52	55.45	55.48	1100	80
7	44.10	44.17	44.13	55.90	55.83	55.98	1200	40
8	43.58	43.64	43.59	56.42	56.36	56.42	1100	40

Table 4
PA and DFT results of the $(1 \pm 2s)f$ sideband frequencies using the minimum window lengths with 1000 Hz sampling frequency for the IM operating under 50% load condition and with various numbers of broken rotor bars.

50% load	$(1-2s)f$ (Hz)			$(1+2s)f$ (Hz)			Min. window length (number of samples)	
Number of broken bars	True value	DFT	PA	True value	DFT	PA	DFT	PA
1	47.16	47.14	47.26	52.84	52.86	52.98	1400	260
2	47.06	46.92	46.91	52.94	53.08	52.99	1300	280
3	46.94	46.92	46.96	53.06	53.08	53.12	1300	280
4	46.79	46.67	46.85	53.21	53.33	53.23	1200	240
5	46.63	46.67	46.59	53.37	53.33	53.39	1200	220
6	46.44	46.36	46.51	53.56	53.64	53.47	1100	210
7	46.20	46.15	46.15	53.80	53.85	53.85	1300	200
8	45.89	46.00	45.89	54.11	54.00	54.16	1000	120

Table 5
PA and DFT results of the $(1 \pm 2s)f$ sideband frequencies using the minimum window lengths with 1000 Hz sampling frequency for the IM operating under 25% load condition and with various numbers of broken rotor bars.

25% load	$(1-2s)f$ (Hz)			$(1+2s)f$ (Hz)			Min. window length (number of samples)	
Number of broken bars	True value	DFT	PA	True value	DFT	PA	DFT	PA
1	48.61	48.52	48.53	51.39	51.48	51.37	2700	500
2	48.56	48.46	48.55	51.44	51.54	51.56	2600	460
3	48.50	48.40	48.49	51.50	51.60	51.44	2500	300
4	48.44	48.33	48.59	51.56	51.67	51.55	2400	250
5	48.34	48.26	48.25	51.65	51.74	51.62	2300	250
6	48.27	48.18	48.27	51.73	51.82	51.75	2200	230
7	48.13	48.10	48.15	51.87	51.90	51.87	2100	230
8	47.99	47.89	48.02	52.01	52.11	51.91	1900	220

This is because the amplitudes of the sideband frequencies are higher when the fault is severer. This makes it easier for PA to estimate their values.

Tables 2–5 are plotted together in Fig. 10 to illuminate the impact of load and the number of broken rotor bars on the minimum length requirements of data windows

for both PA and DFT. It should also be noticed that the data acquisition time is in direct proportion to data window length when the sampling frequency is determined. Therefore, Fig. 10 reveals that, in all circumstances those have been taken into account, PA needs a considerably shorter data window (or data acquisition time) for distinguishing the two sideband frequency components compared to DFT. It also can be observed that with the same fault severity, when the machine load decreases, which means the two broken bar sideband frequency components will move closer to the fundamental frequency, the minimum window length required for DFT increases dramatically, whereas that for PA only raises sightly. The trends of the minimum window length requirements of the two methods illustrate that if higher frequency resolutions are needed, the data window (or data acquisition time) for DFT has to be enlarged (or lengthened) significantly to observe close frequencies, whereas the PA only needs slightly longer data windows (or data acquisition time). In practice, this advantage of PA can constrain the effects of changing sideband frequencies caused by time-varying machine load.

4.2.2. Accuracy

In practice, more precise estimates of the values of the broken rotor bar sideband frequencies facilitate decisions on the existence of broken rotor bars to be more creditable. The accuracy of the frequency estimation by DFT depends on the frequency resolution, which is solely determined by the sampling time. The accuracy of the frequency estimation by PA is also affected by the window length. In this section, they are compared in terms of the unitary Mean Absolute Error of the frequency estimation, which is calculated as the mean of the unitized absolute error of frequency estimates for a number of independent runs. The equation is given as

$$MAE_{freq} = \frac{\sum_{n_t=1}^{N_t} UE_{n_t}}{100N_t} \tag{20}$$

where N_t is the number of independent trials.

Figs. 11 and 12 show the accuracy of frequency estimation in terms of MAE_{freq} for both PA and DFT. The MAE_{freq} is calculated for 100 independent trials with each window size, varying from 500 samples to 3500 samples using 1000 Hz sampling frequency. To construct the independent trials for testing the two spectral analysis methods, 10 independent simulation runs, each of which lasts for 40 s, with random selections of errors are executed. A different segment of data is used for each trial. The stator current signals are obtained by simulating a 5.5 kW, four pole induction machine with the number of broken rotor bars varying from 1 to 8. The full load and 75% conditions are used for these figures.

It is observed that although the MAE_{freq} of DFT decreases when the data window is enlarged, in general, much better estimation results obtained by PA using a window of the same length is achieved. With a range of different window lengths taken into account, the comparison result highlights that PA is capable of estimating the sideband

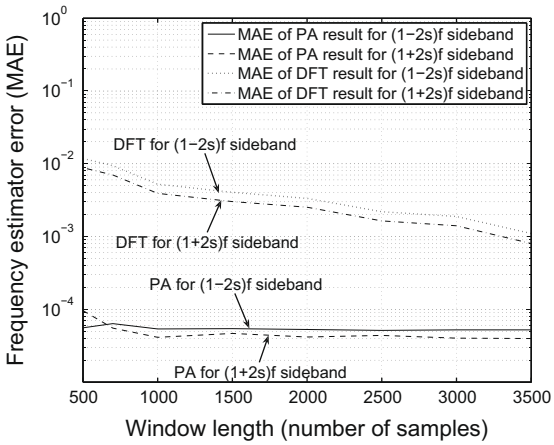

Fig. 11. MAE_{freq} in frequency estimation by PA and DFT in respect of window length using 1000 Hz sampling frequency for simulated current data of the IM operating under full load.

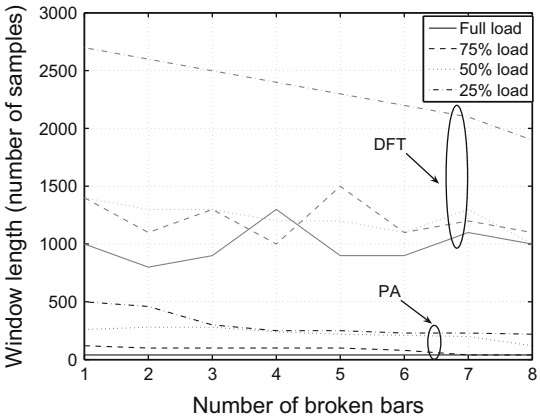

Fig. 10. Plotted comparison of the minimum window length requirements of PA and DFT for broken rotor bar detection with respect to different numbers of broken rotor bars and load conditions.

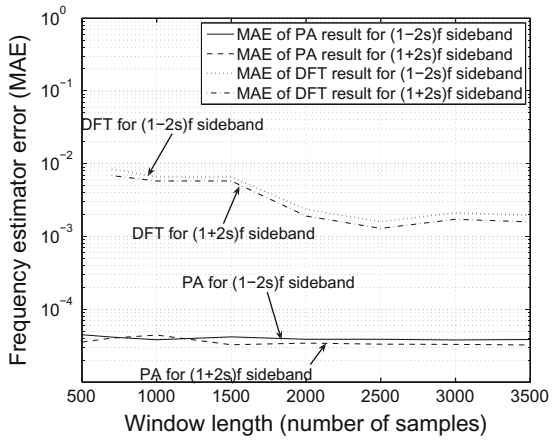

Fig. 12. MAE_{freq} in frequency estimation by PA and DFT in respect of window length using 1000 Hz sampling frequency for simulated current data of the IM operating under 75% of full load.

frequency components more accurately than DFT. It is also observed that the MAE_{freq} remains in almost the same level for PA regardless of the window length, though there is an approximate minimum window length required for it to be able to function, as described previously.

The window length can be critical for on-line diagnosis of induction machine broken rotor bars. The effect of light or changing load is desired to be eliminated as much as possible. Therefore the requirement of long data windows for DFT makes it very difficult to detect the sideband frequencies in such conditions.

4.2.3. Light load conditions

As mentioned that the broken rotor bar sidebands are very close to the fundamental frequency when the machine load is light, another interesting fact is that what the lightest machine load is, for PA to be able to accurately estimate the two sideband frequencies. Table 6 displays the true and estimated values of sideband frequency components obtained by PA in regard to conditions that the load is less that 25% of the full load. The sampling frequency is 1000 Hz.

Table 6 shows that the PA has successfully estimated the two broken bar sideband frequencies when only one broken rotor bar exists and the load is as light as only 1% of full load. This is an extreme case as it is required to distinguish considerably small peak in the vicinity of a much higher frequency peak. The amplitude of the two sideband frequencies in the 1% load case, are −65.33 dB and −71.77 dB lower than the fundamental frequency, respectively, due to the extremely light load. The frequency difference between the nearest high peak is only 0.05 Hz. These would make the two frequency components almost enshrouded in the noise floor or the side lobe of the fundamental frequency due to spectral leakage, if the DFT is used. With the same load condition, the one broken rotor bar case can be considered as the most difficult one. Thus, it is reasonable to state that if more than one broken bar exist in the rotor, they also can be detected. To achieve the similar resolution, DFT requires much longer data windows, as presented, which is obviously inconvenient and difficult in practice.

Since the small value of motor slip when the load is light, more sideband harmonics will also appear very close to the fundamental frequency if the number of broken rotor bars increases. This also increases the complexity of the computation for PA. Therefore, a filter of narrower passband is used to eliminate those interfering frequency components. Here, the filter used for small load conditions has the frequency passband of $f \pm 3$ Hz, passing only the fundamental frequency and the two sideband frequencies.

5. Experiment tests

5.1. Experiment setup

In order to verify the PA method for practical uses, measured data from laboratory experiments is used in this section. A commercial 2.2 kW, 50 Hz, four poles induction machine, which has a standard cast aluminum squirrel cage rotor with 32 rotor slots, is used in the test. A separately-excited DC generator is coupled via a belt as load, and is loaded by using a variable resistance bank. The broken rotor bar fault is constructed by cutting holes through the rotor bars at the joints with the end ring using a fine milling cutter.

The stator current is sensed by a Hall-Effect clamp probe, passed through an anti-aliasing filter, which is an 8th order Butterworth filter, and finally sampled by an A/D converter with a sampling frequency of 400 Hz. This gives a Nyquist frequency of 200 Hz. The sampling time is 20s, which gives the longest window size of 8000 samples. A custom written LabVIEW data acquisition system is used for data acquisition [22].

The captured data was then analyzed using both PA and DFT in Matlab. The data is passed through a Hanning window before applying DFT, and is passed through an FIR bandpass filter, which is the one mentioned in Section 3.1, before applying PA.

5.2. Test results

Four bars are cut in the rotor and the induction motor is managed to operate with full load. The spectrum of the measured stator current using DFT is presented in Fig. 13. The two sideband frequency components at 43.30 Hz and 56.70 Hz and their harmonics can also be observed. The whole data is used for DFT, giving the frequency resolution determined as 0.05 Hz.

In this implementation a remarkably short data window of only 200 samples is used for PA. The order of the algorithm is chosen as six since the number of frequency components is known. The PA result is displayed in Fig. 14 and Table 7. The rotor speed is measured using a digital photo tachometer to calculate the slip and the true value of sideband frequencies.

Table 6
Estimated values of the sideband frequency components by PA and DFT for broken rotor bar detection under different light load conditions and with one broken rotor bar.

Load condition	$(1-2s)f$ (Hz)			$(1+2s)f$ (Hz)			Min. window length (number of samples)	
(% Of full load)	True value	DFT	PA	True value	DFT	PA	DFT	PA
20	48.90	49.00	48.59	51.10	51.00	51.66	3000	100
15	49.17	49.07	49.17	50.83	50.93	51.65	4000	200
10	49.45	49.50	49.32	50.55	50.50	50.41	6000	1000
5	49.73	49.75	49.43	50.27	50.25	50.18	12,000	2000
1	49.95	49.95	49.71	50.05	50.05	50.89	35,000	2000

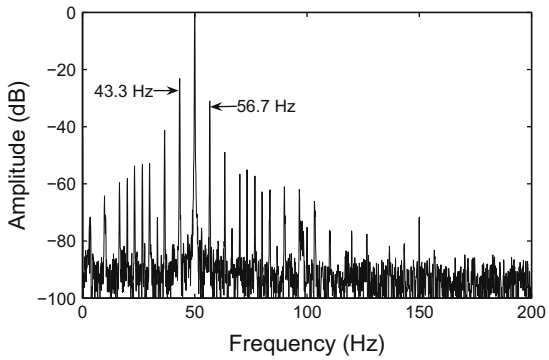

Fig. 13. Spectrum of the measured stator current of a 2.2 kW induction motor with four broken rotor bars using DFT with a sampling frequency of 400 Hz, and a data window of 4000 samples.

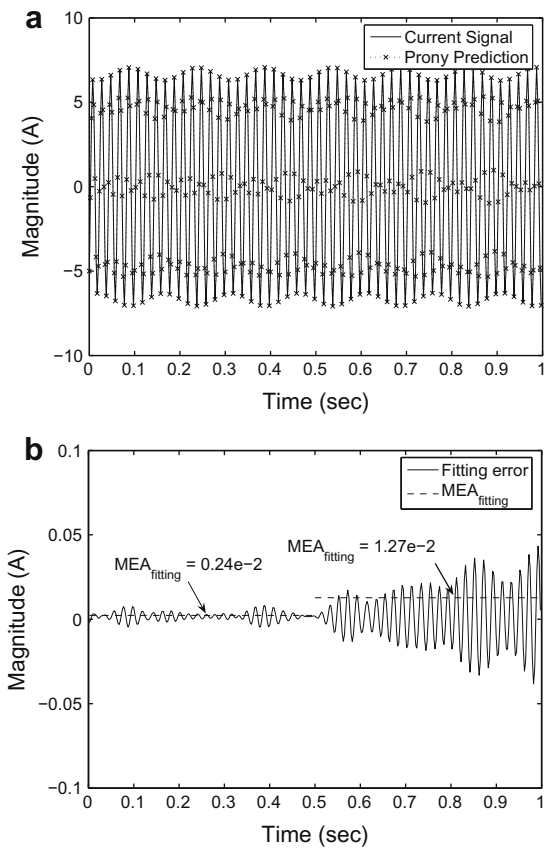

Fig. 14. Comparison of the current waveforms between PA prediction with the real current signal inside and outside the window of a 2.2 kW induction machine operating in full load with four broken rotor bars. (a) PA prediction result inside and immediate after the window. (b) Difference between the current signal and the PA prediction.

The plotted PA prediction result of the current signal both inside and outside the window in Fig. 14 demonstrates an excellent match with the measured real current waveform. The $MAE_{fitting}$ of the estimation part of signal is 0.0024 and the $MAE_{fitting}$ of the prediction part of signal is 0.0127. The numerical result in Table 7 shows a high resolution achieved by PA, while the DFT result presents big errors in determining the frequency values. The spectral estimation result obtained by DFT using 200 data samples is also plotted in Fig. 15 to compare against the result of PA. The comparison clearly shows the superiority of the PA.

6. Evaluation of Prony Analysis

The IRLS Prony method is a significant improvement of the Prony algorithm based on LS Prony and has shown a strong ability of dealing with noisy signals in practice. It is therefore the method chosen to be utilized in this research. The purposes of this section are to try to characterize the influential factors and to have a better understanding of how to adjust them to make PA work better. The evaluation of PA is completed by introducing influencing factors and examining their impacts on the frequency estimation accuracy. The factors which are discussed are the data window length and the noise level.

The success of Prony fitting and frequency estimation is subject to a number of influencing factors. Only experiential conclusions have been made so far on how these factors affect Prony fitting. Signal noise level, amplitude of each frequency component contained in the signal, window length, sampling frequency, algorithm order choice, the number of samples taken into the estimation process or even the data samples chosen all affect the estimation result. This is both reported in [21] and observed by the authors.

6.1. Impact of data window length

It has been shown in Section 4.2.2 that if the length of data window for PA is more than 500 samples with 1000 Hz sampling frequency, even if the window length is enlarged further, the accuracy of frequency estimation remains in a similar level. However, if a considerably small data window is used, the performance of PA is effected greatly by the length of the window.

Fig. 16 displays the MAE_{freq} of the higher and lower sideband frequency estimations with respect to the length of the data window less than 500 samples using 1000 Hz sampling frequency. The data used is the simulated stator current signal of the induction machine model used in former sections. The number of broken rotor bars is 2 and the load condition is full load. The MAE_{freq} is calculated using Eq. (20) for 100 independent trials.

It is observed that if only a little data is available the estimator error can be much higher. However, after the amount of data samples used in the Prony algorithm reaching to a certain number, the frequency estimator error falls down and then remains almost steady even the data window length continues to increase. The length of the data window at this turning point of the MAE_{freq} curve is the minimum window length requirement. The estimator error is generally higher for the $(1+2s)f$ sideband frequency. This is due to that its magnitude is generally lower than that of the $(1-2s)f$ sideband frequency.

Table 7
PA result of the measured stator current signal of a 2.2 kW induction motor with four broken rotor bars operating under full load condition, using a data window of 200 samples with the sampling frequency of 400 Hz.

	True value (Hz)	PA		DFT	
		Value (Hz)	Amplitude (dB)	Value (Hz)	Amplitude (dB)
$(1-2s)f$	43.33	43.30	−22.75	44.00	−23.88
Fundamental	50.00	49.99	0.00	50.00	0.00
$(1+2s)f$	56.67	56.68	−30.63	56.00	−31.40

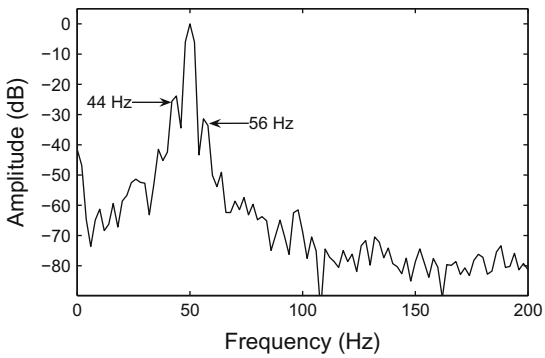

Fig. 15. DFT spectrum of the same signal data used in Figs. 13 and 14 but using a window of only 200 samples with the sampling frequency of 400 Hz.

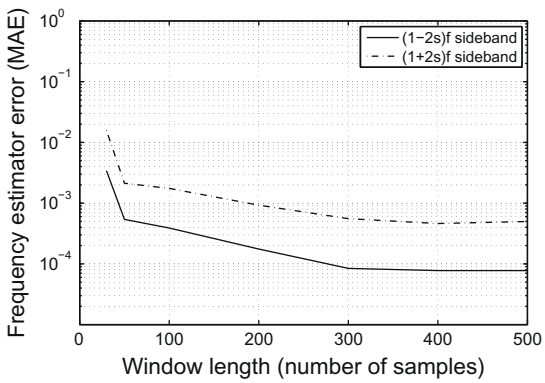

Fig. 16. MAE_{freq} of the six order PA frequency estimator of broken rotor bar sideband frequencies with respect to the window length when using 1000 Hz sampling frequency for 100 runs.

6.2. Impact of noise

Noise presented in the signal can deteriorate the performance of PA. The $MAE_{fitting}$ of Prony estimation calculated using Eq. (18) is displayed in Fig. 17. The signal is sampled with a sampling frequency of 1000 Hz and the data window is 500 samples. The standard deviation of the measurement error is simulated by using the random number generator. It is increased from 0.005 to 0.105 with a step of 0.01. The results clearly show the trend of the degree of accuracy that IRLS Prony is able to perform in modeling the original signal. A higher noise level decreases the accuracy of estimate.

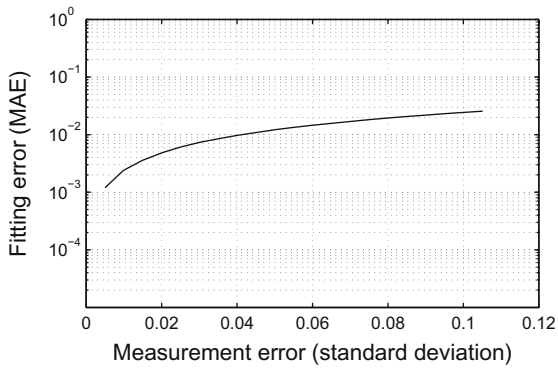

Fig. 17. Estimation mean absolute error as a function of measurement error standard deviation for IRLS Prony.

7. Conclusion

This paper has proposed PA in the diagnostics of induction motor broken rotor bars. The PA is not only able to estimate the data within the data window, but also predict the future data with high precision. Compared with DFT, the PA has demonstrated great advantages in terms of using much shorter data windows to satisfy the same frequency resolution requirement, and gaining a higher accuracy of frequency estimation using the same length windows. This gives the PA the ability to detect the broken rotor bar sideband frequency components in light and time-varying load conditions. The minimum window length required by DFT can be 6–20 times longer than PA in light load conditions. Moreover, it is also possible to extend this method to diagnose other types of motor faults using spectral analysis based MCSA.

Data pre-conditioning such as filtering, downsampling and constant offset removing needed to be conducted in DSP devices prior to running the PA algorithm in on-line diagnostics applications.

Evaluation results show that the PA algorithm needs a minimum length window to function correctly. If this minimum window length is not achieved, PA may fail. However, the accuracy of frequency estimation using PA shows great independence on the data window length once the minimum length requirement has been met. The noise level also affects the accuracy of the PA result. A higher noise level will result in a lower estimate accuracy.

The order of the Prony algorithm is always chosen as twice the number of frequency components present in the signal. Thus, previous knowledge of the number of signal poles is desired. Besides, since the PA is a high

computational cost algorithm, the order is preferred to be as small as possible.

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